

Session 17: Introduction to JavaScript & Basics

Session Objective

Introduce learners to **JavaScript fundamentals**, set up a working environment, and help them write and run basic programs using variables, data types, and operators.

Duration

3 Hours

Instructor Guidelines

- Follow Agenda: Please adhere to the provided session agenda to ensure effective and consistent training delivery. Use the provided training resources such as slide decks and code samples to teach in the class. Use references, wherever provided to learn and prepare with factual content before conducting the session.
- Check Audio Equipment: Verify that your microphone and speakers/headset are functioning properly before starting the session. Please use a good quality microphone and completely avoid using bluetooth earbuds as they employ noise-cancellation which reduces audio quality.
- Camera Usage: Please ensure your camera is on for at least the first 15 minutes of the session.
- Power Backup: Please ensure you have a power backup solution ready to prevent interruptions due to power outages.
- Internet Bandwidth: Please confirm that your internet connection is fast enough to handle live-streaming audio, video, and screen sharing.
- Code/Query Testing: Run through your code/queries before the session to prevent any disruptions.
- Hands-On Coding: During interactive coding segments, explain steps clearly and frequently check in with students to confirm their understanding.
- Technical Support: If you encounter issues with Zoom, contact the session moderator or the learning advisor for help.

Agenda

1. Session Opening

- Greet learners and take attendance
- Transition from HTML/CSS → JavaScript
- Explain today's goal: "Make web pages dynamic using JS"

- Ask learners to open VS Code

2. What is JavaScript

- Explain JavaScript as a programming language for logic and interactivity
- Clarify where JS runs:
 - browser (frontend)
 - Node.js (backend/runtime)
- Give simple examples:
 - button click
 - form validation
- Keep conceptual, not deep

3. Node.js Setup & Running JS

- Check Node installation:
 - `node -v`
- Guide installation if missing
- Create a file: `app.js`
- Run using:
 - `node app.js`
- Show simple output:
 - `console.log("Hello World")`
- Confirm all learners can run JS

4. Values and Data Types

- Explain what a value is
- Introduce basic data types:
 - number
 - string
 - boolean
 - null (mention)
 - undefined (mention)
- Demonstrate using console:
 - `console.log(typeof value)`
- Keep examples simple

5. Variables (let, const, var)

- Explain purpose of variables (store data)
- Introduce:
 - `let` (changeable)
 - `const` (fixed)

- `var` (avoid, legacy mention)
- Show examples:
 - declare → assign → print
- Emphasize:
 - prefer `let` and `const`

6. Identifiers & Naming Rules

- Explain identifiers as variable names
- Rules:
 - cannot start with number
 - no spaces
 - case-sensitive
- Show good vs bad examples
- Emphasize readable naming

7. Comments

- Explain purpose of comments
- Demonstrate:
 - single-line (`//`)
 - multi-line (`/* */`)
- Encourage use for clarity

8. Break

- Ask learners to step away, hydrate, stretch
- Resume on time

9. Basic Operators

- Introduce arithmetic operators:
 - `+`, `-`, `*`, `/`, `%`
- Introduce comparison operators:
 - `==`, `===`, `!=`, `>`, `<`
- Demonstrate using console outputs
- Highlight difference between `==` and `===` (basic only)

10. Practice Exercises (Hands-On)

- Ask learners to:
 - declare variables and print values
 - perform arithmetic operations
 - compare values and print results
- Give 2–3 small tasks (quick checks)

11. Doubt Resolution

- Address:
 - let vs const confusion
 - data types confusion
 - node not running
 - operator behavior